A FRIENDLY INTRODUCTION TO COSMOLOGY

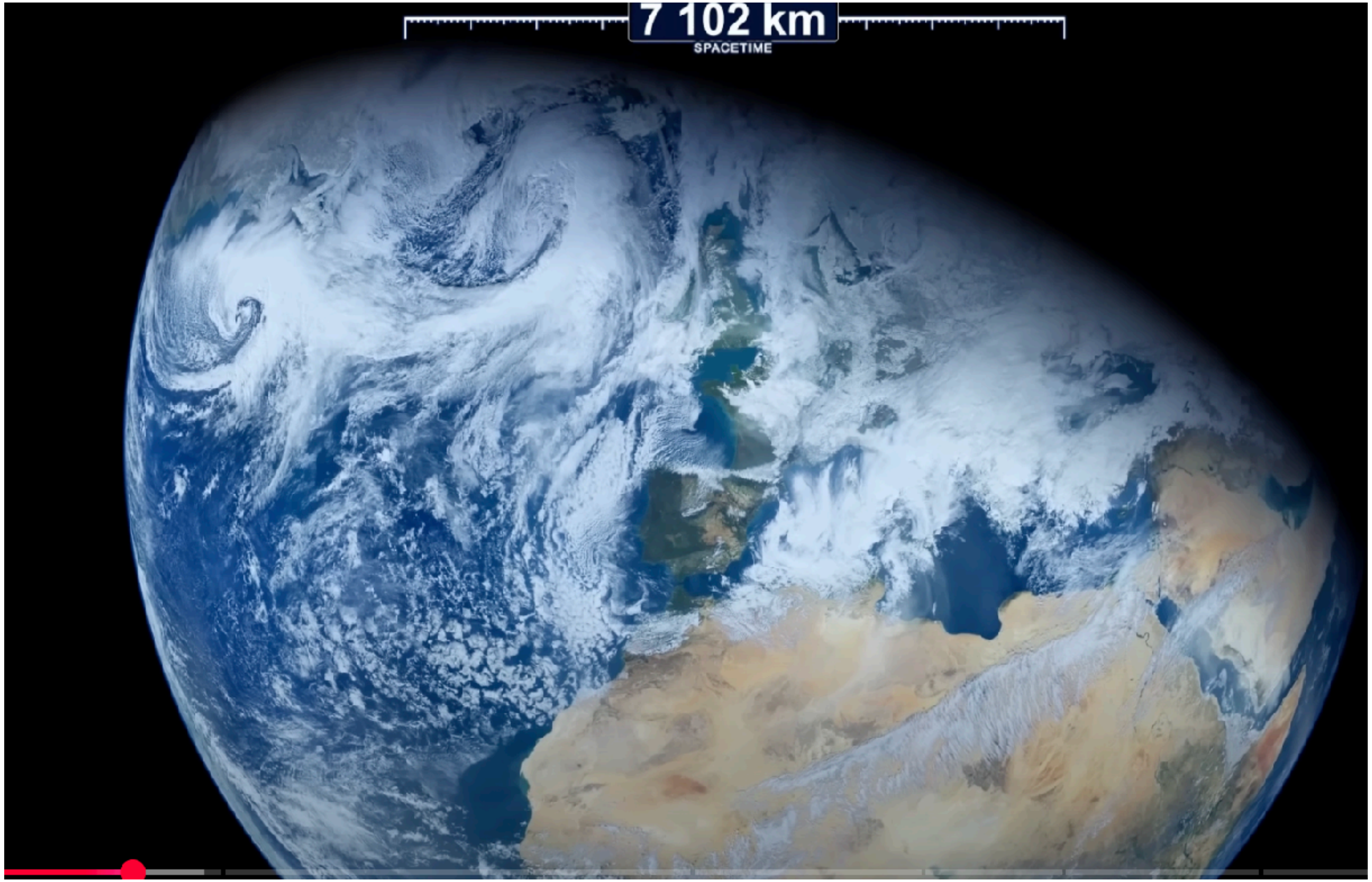
RENAN ISQUIERDO BOSCHETTI

WHY IS COSMOLOGY INTERESTING?

WHAT IS COSMOLOGY? NOT ABOUT YOUR HOUSE

Credit: https://www.youtube.com/watch?v=U9WOMerccoM&t=560s

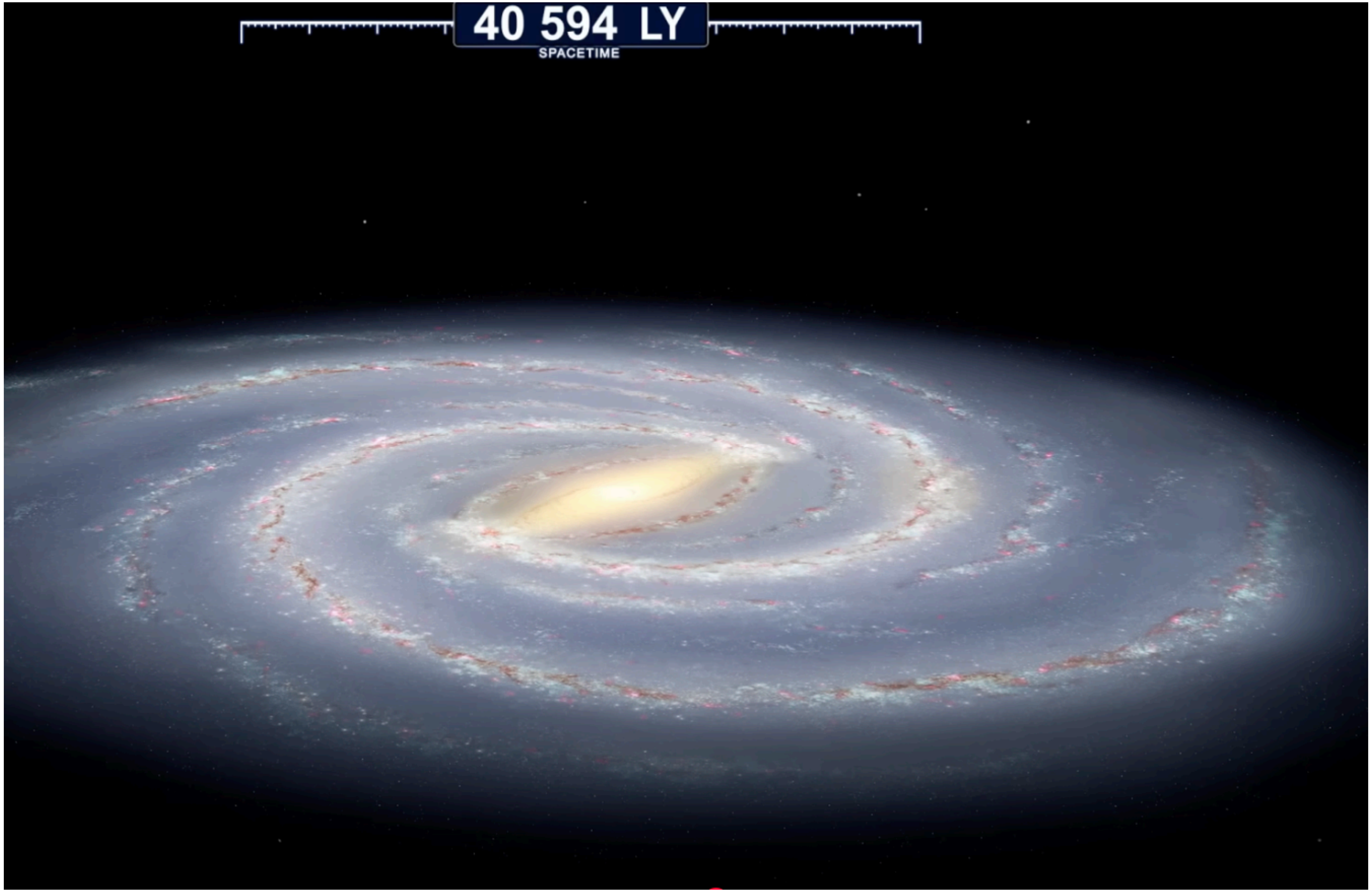
WHAT IS COSMOLOGY? NOT ABOUT YOUR RIDICULOUSLY SMALL PLANET



WHAT IS COSMOLOGY? NOT ABOUT YOUR INSIGNIFICANT SOLAR SYSTEM

WHAT IS COSMOLOGY? NOT ABOUT STARS

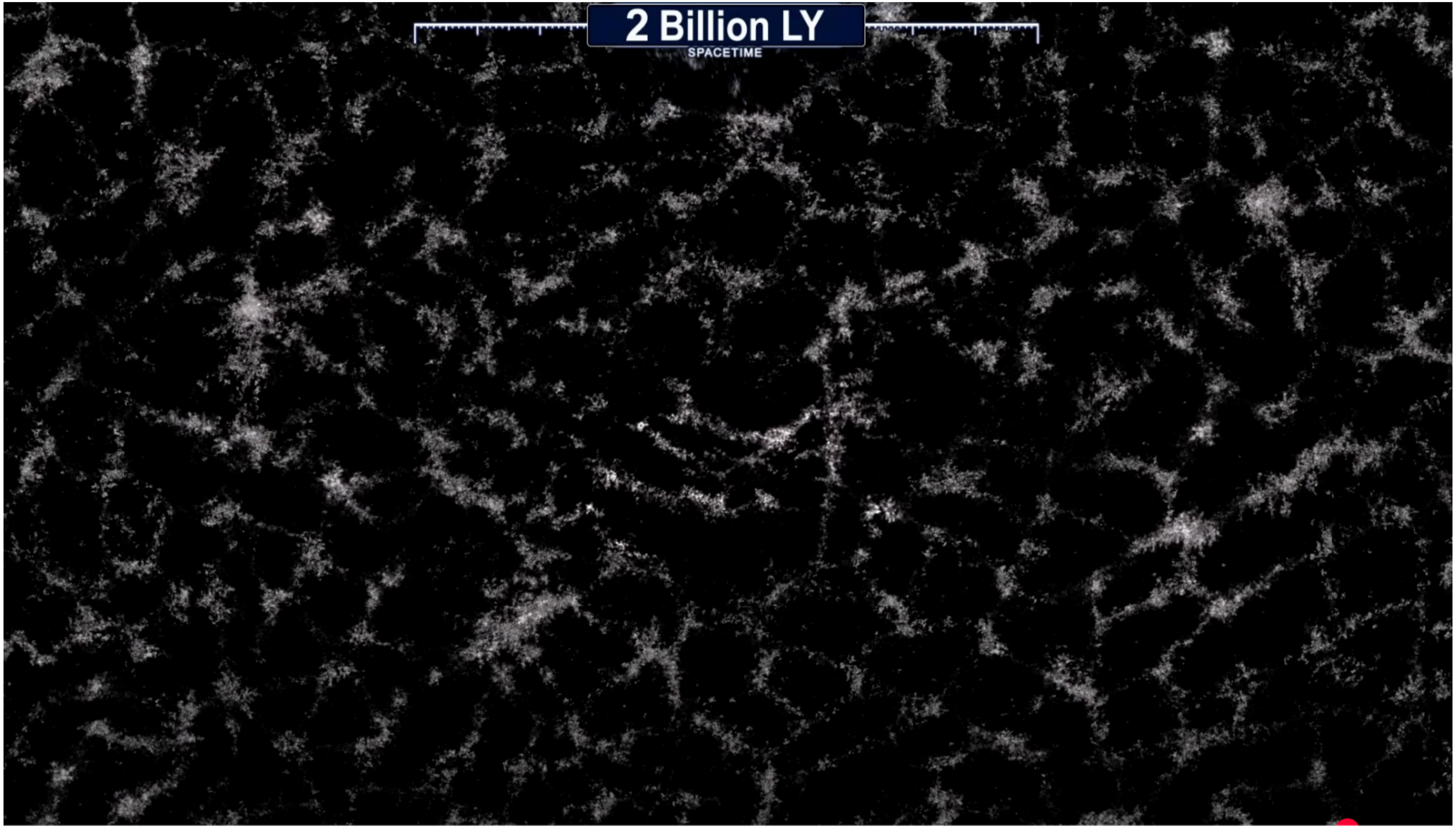
WHAT IS COSMOLOGY? NOT QUITE THERE YET



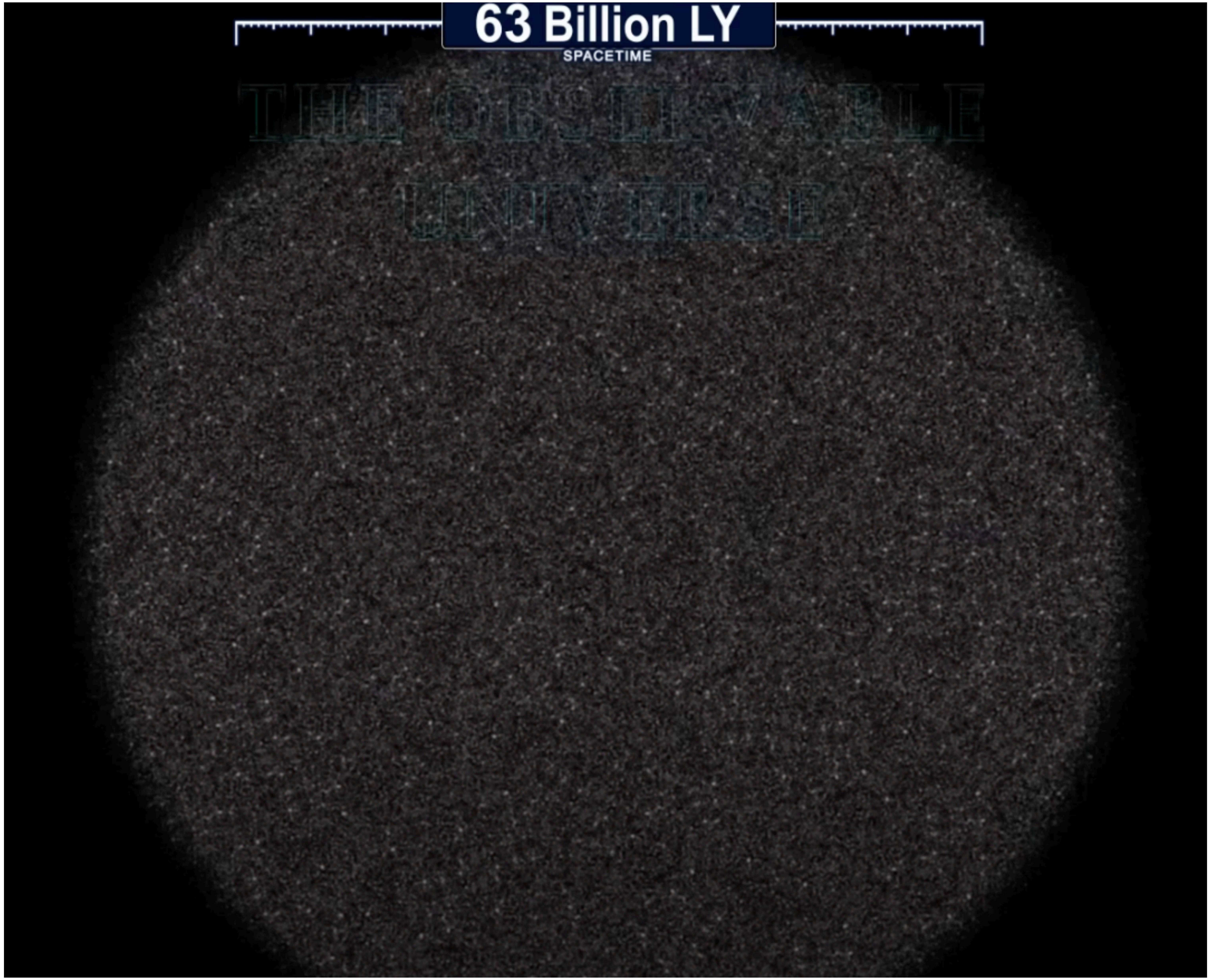
WHAT IS COSMOLOGY? NOW WE START TALKING

WHAT IS COSMOLOGY? NOW WE CAN THINK ABOUT COSMOLOGY

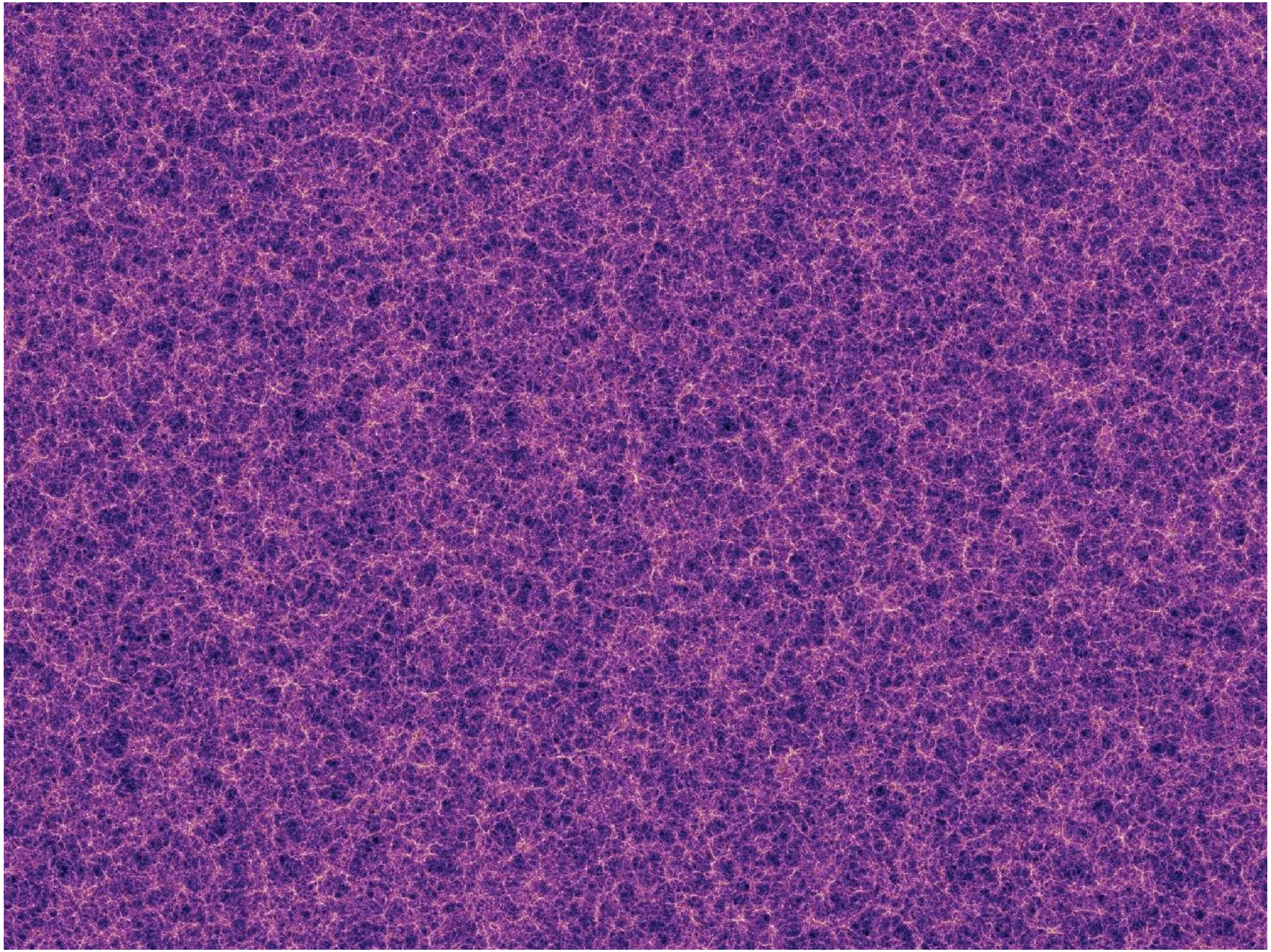
WHAT IS COSMOLOGY? NOW WE CAN ACTUALLY DO COSMOLOGY



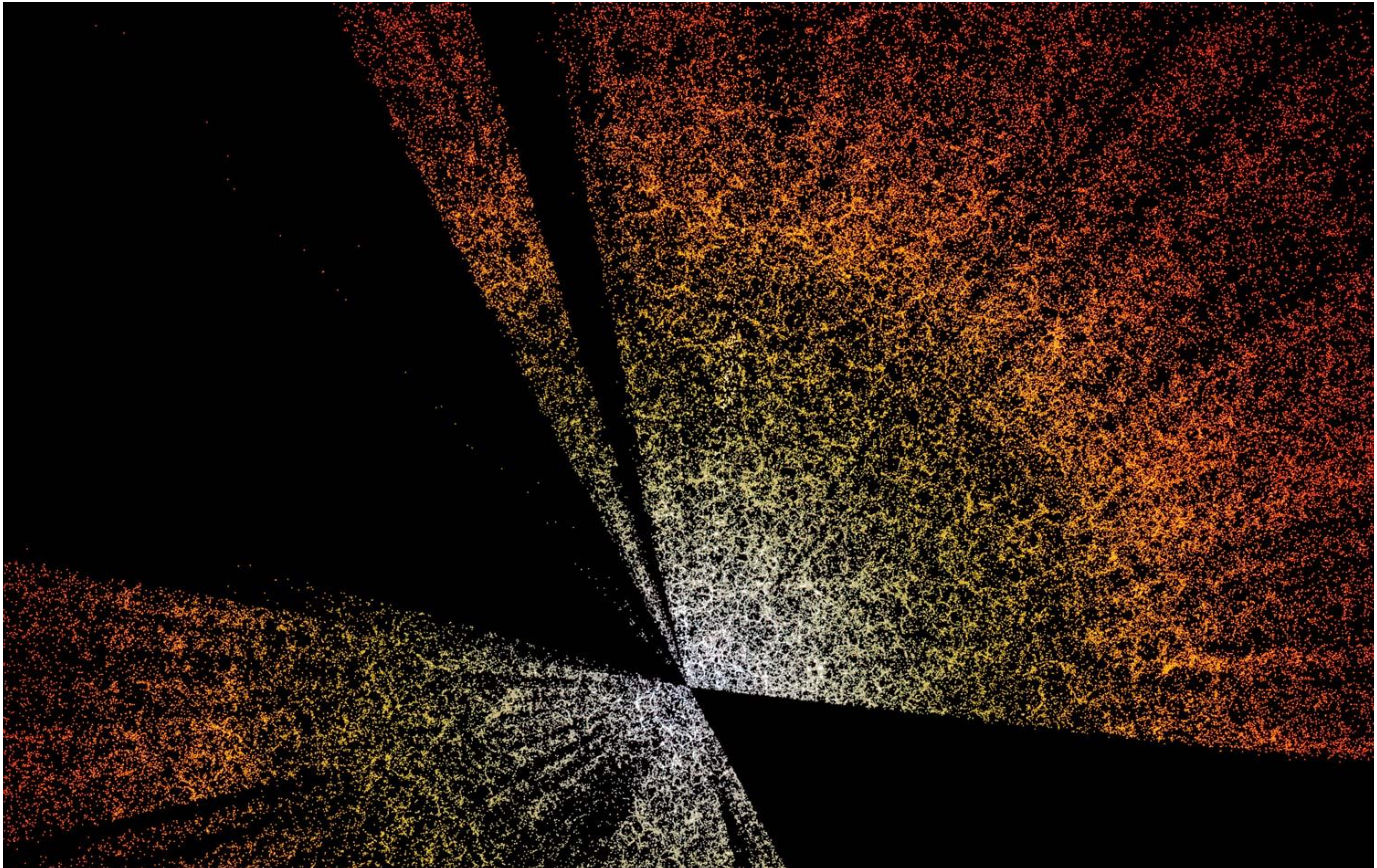
WHAT IS COSMOLOGY? EVERYTHING WE CAN EVER KNOW ABOUT COSMOLOGY



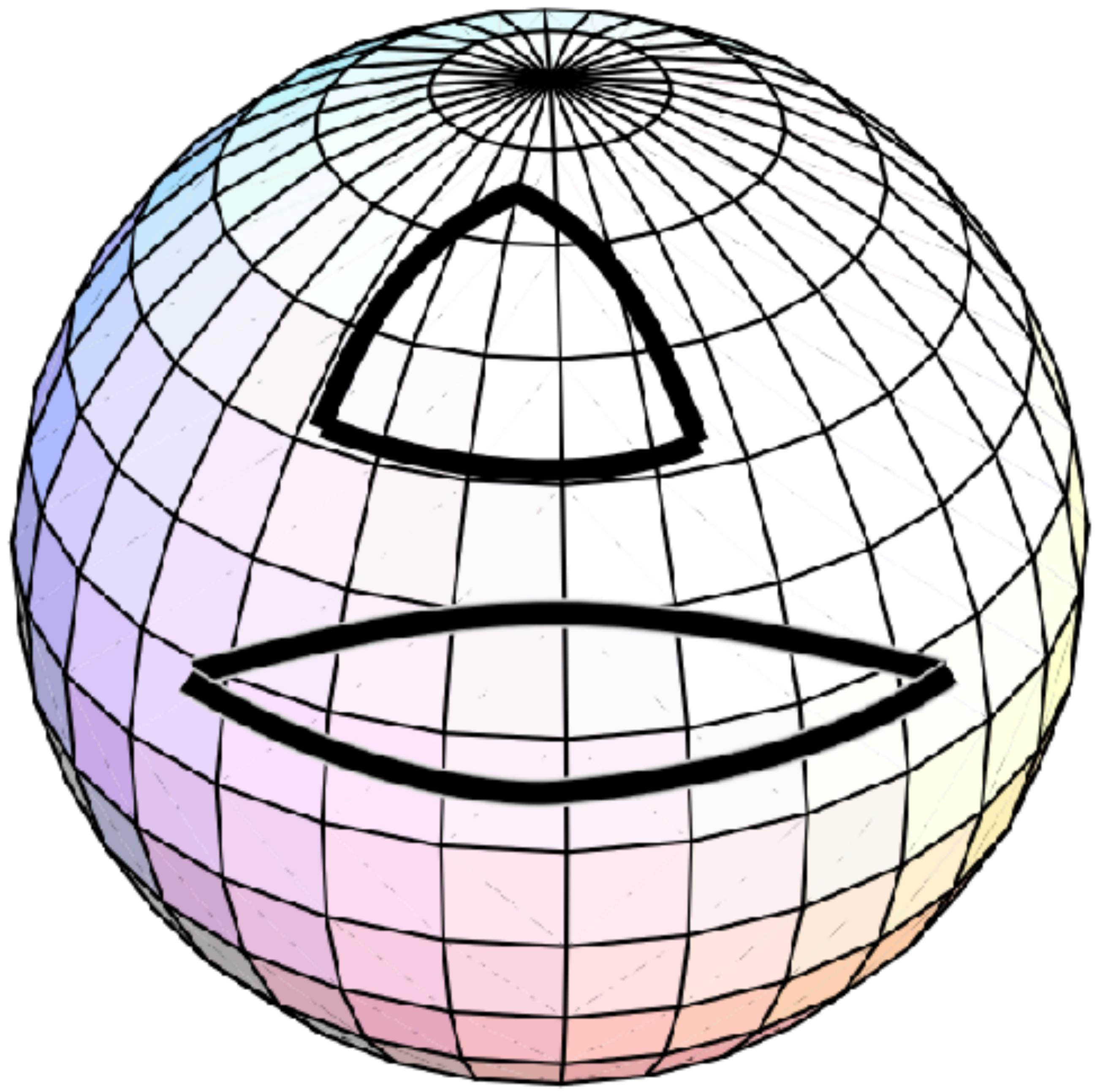
THE COSMOLOGICAL PRINCIPLE (CREDIT: MILLENNIUM SIMULATION)



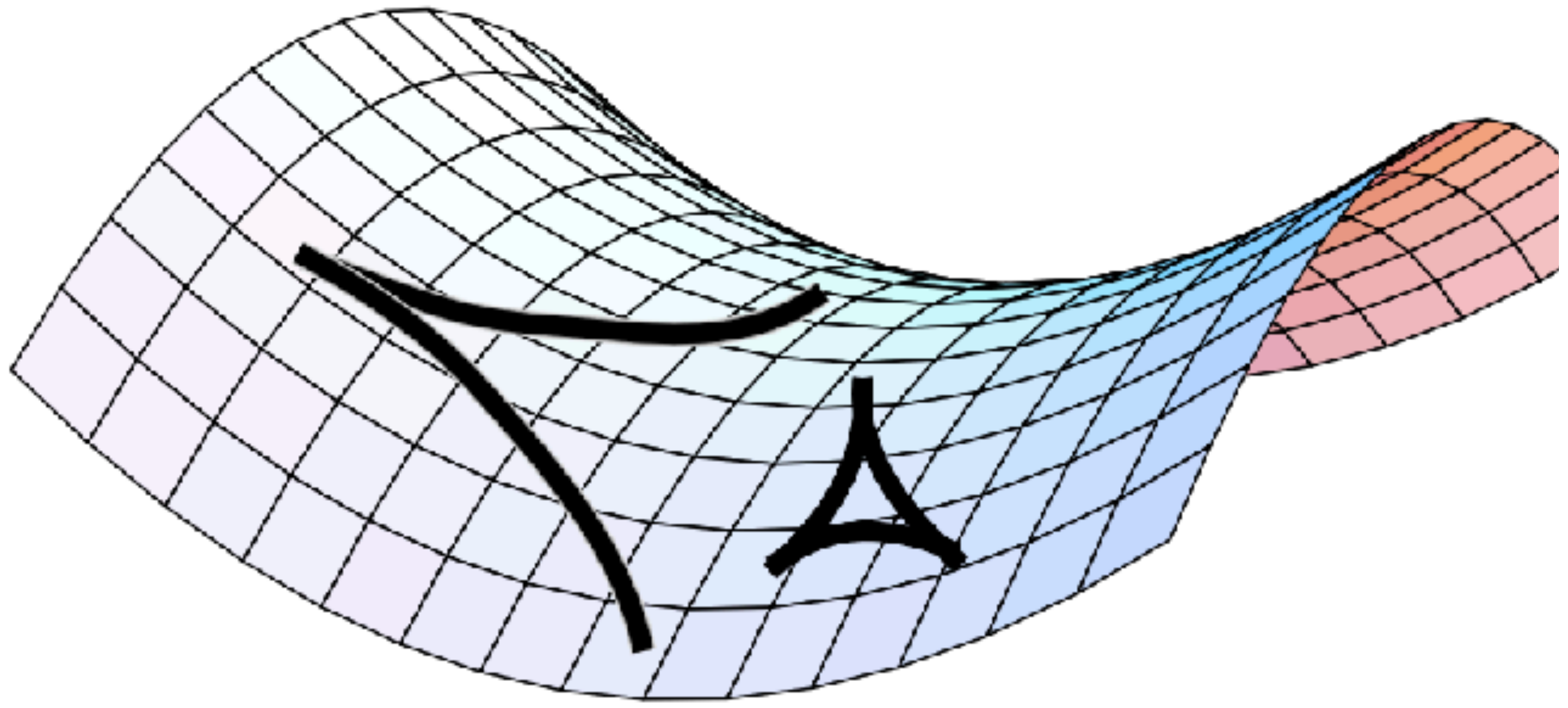
WHAT IS COSMOLOGY? GALAXY SURVEYS (CREDIT: DESI COLLABORATION)



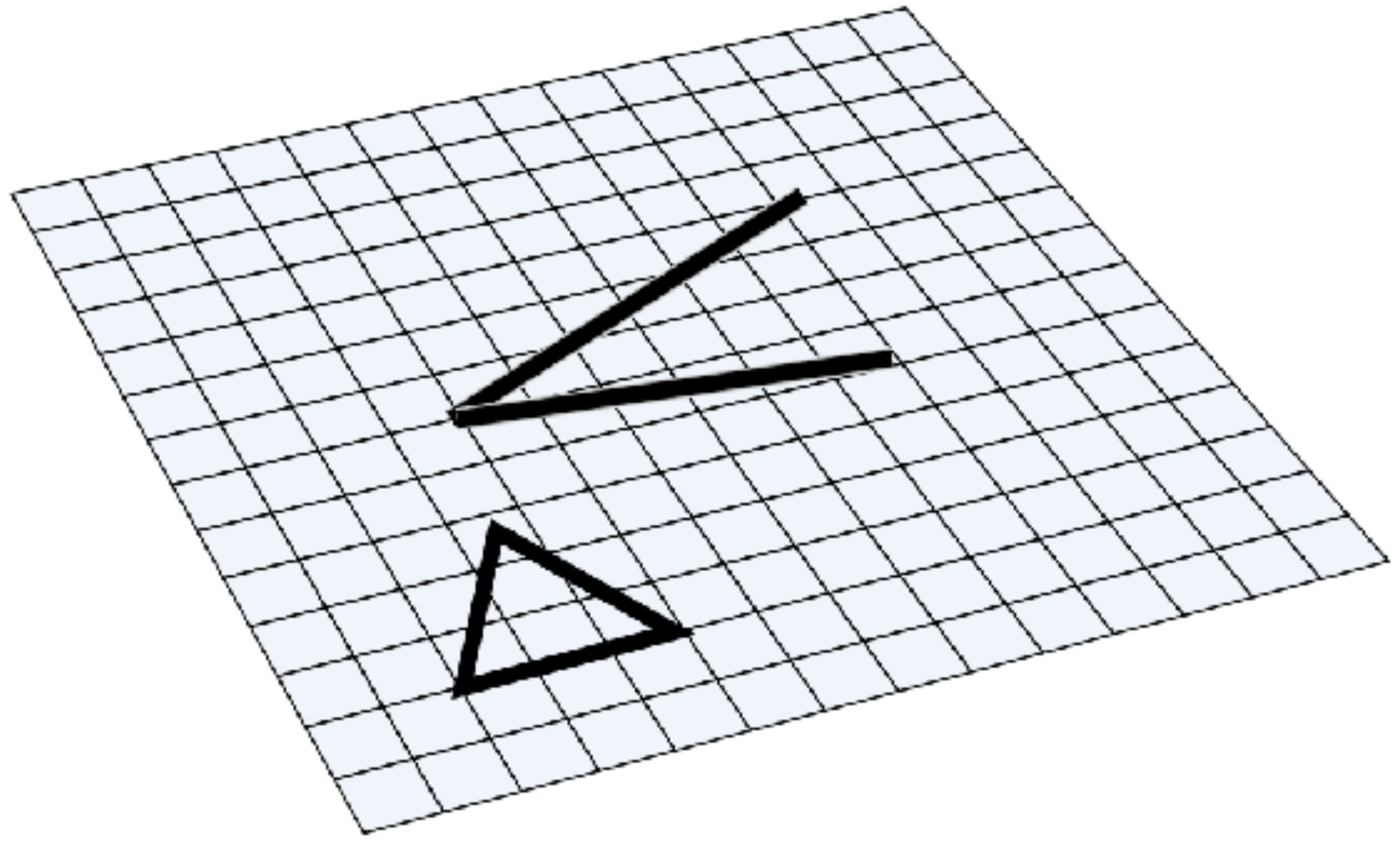
(POSSIBLE CURVATURES OF AN HOMOGENEOUS AND ISOTROPIC UNIVERSE)



Positive curvature

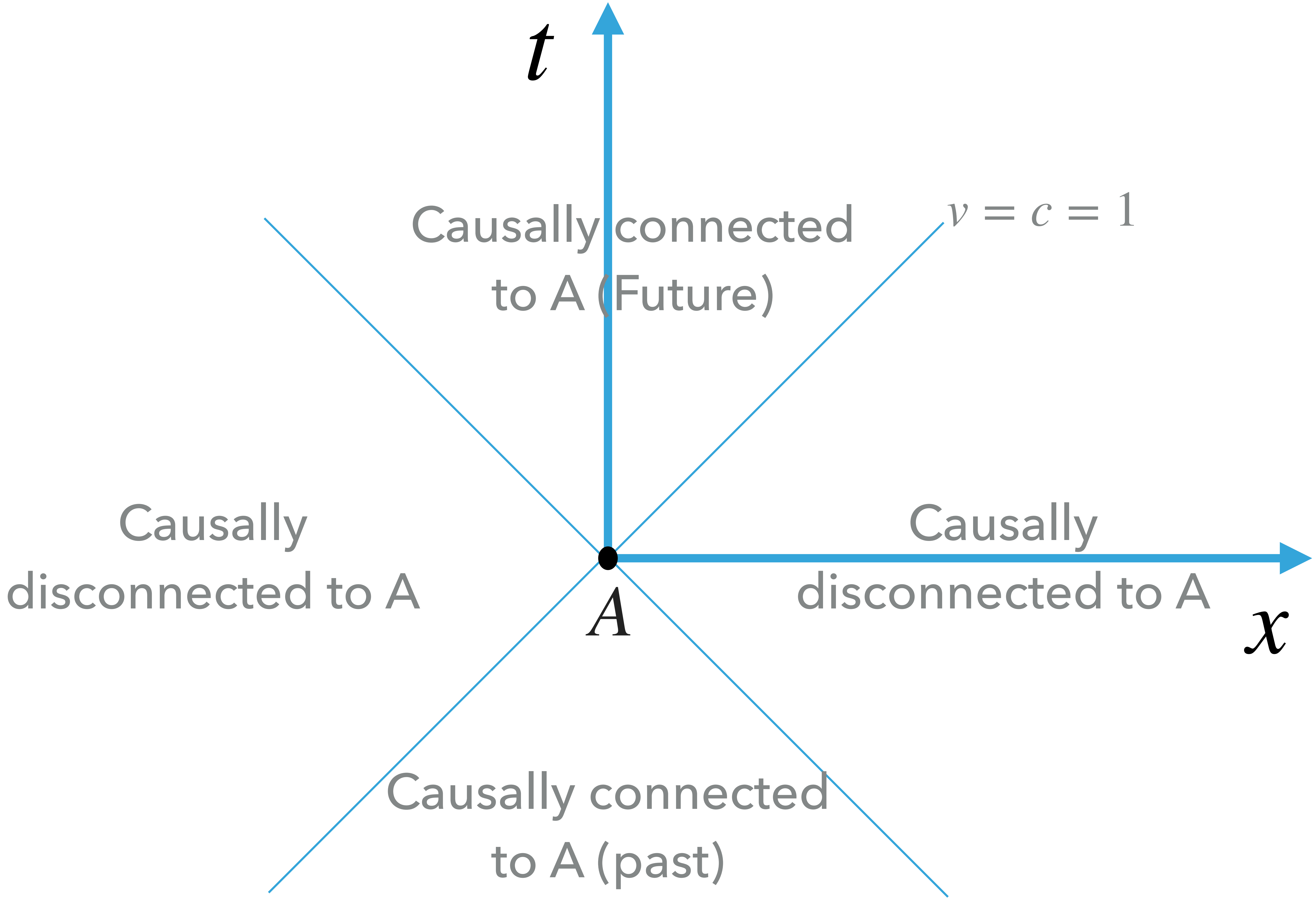


Negative curvature



No curvature

WHAT IS THE UNIVERSE MADE OF – SPACETIME (EINSTEINIAN)



WHAT IS THE UNIVERSE MADE OF - SPACETIME (EINSTEINIAN) - CREDIT: SALVATORE ORLANDO

SUMMARY

- Cosmology is the branch of physics that studies the Universe in its largest scales.
- On the largest scales, the Universe forms structures, as a result of gravitational collapse (and the initial conditions)
- On large scales, the Universe becomes Homogeneous and Isotropic, which means that, statistically, on a sufficient large scale, different patches of the Universe are indistinguishable.
- The Universe has a geometry. Since it is homogeneous and Isotropic, there are three possibilities: flat, closed (positive curvature), and hyperbolical (negative curvature).
- We can only observe a finite region of the Universe. This happens because there is a fundamental limit for how far a particle can travel (the speed of light).
- By looking to distant galaxies, we are also looking to the past, or to our past light cone.